

Programming Language C

Language

❖ **Language:** A language is a structured system for communication.

❖ **Natural Language:** A structured system for communication between two or more human beings. For example, Bangla, English, etc.

❖ **Programming Language:** A structured system of writing programs for either (i) communication between two or more machines; or (ii) human machine interaction.

Constituents of Language

Natural Language	Programming Language
Letter	Character
Alphabet	Character Set
Word	Token
Sentence	Statement
Essay	Program

Character Set

- ❖ Computer can store or recognize only **0** and **1**. Letter or character like A, B can be stored or recognized by a computer. Hence, the characters are represented by some numbers, known as **codes**. For example, 'A' is represented by 65, 'a' is represented by 97, '0' is represented by 48 and so on.
- ❖ In America, 128 characters are coded with the numbers ranges from 0 to 127, which are known as **ASCII** (American standard code for information interchange). Another 128 characters are coded with the numbers ranges from 128 to 255, which are known as **extended ASCII**. All of these 256 characters are supported by C programming. **8 bits or 1 byte** memory is used to represent an ASCII Character.
- ❖ Later on, all the symbols used in worldwide languages are coded using 16 bits or 2 bytes, which is known as **UniCode** (Universal codes). A Maximum of 65536 characters can be coded using Unicode. **C programming does not support Unicode.**
- ❖ Each character is represented within single quotes like 'A', 'B', '0' and so on.

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Escape Sequence

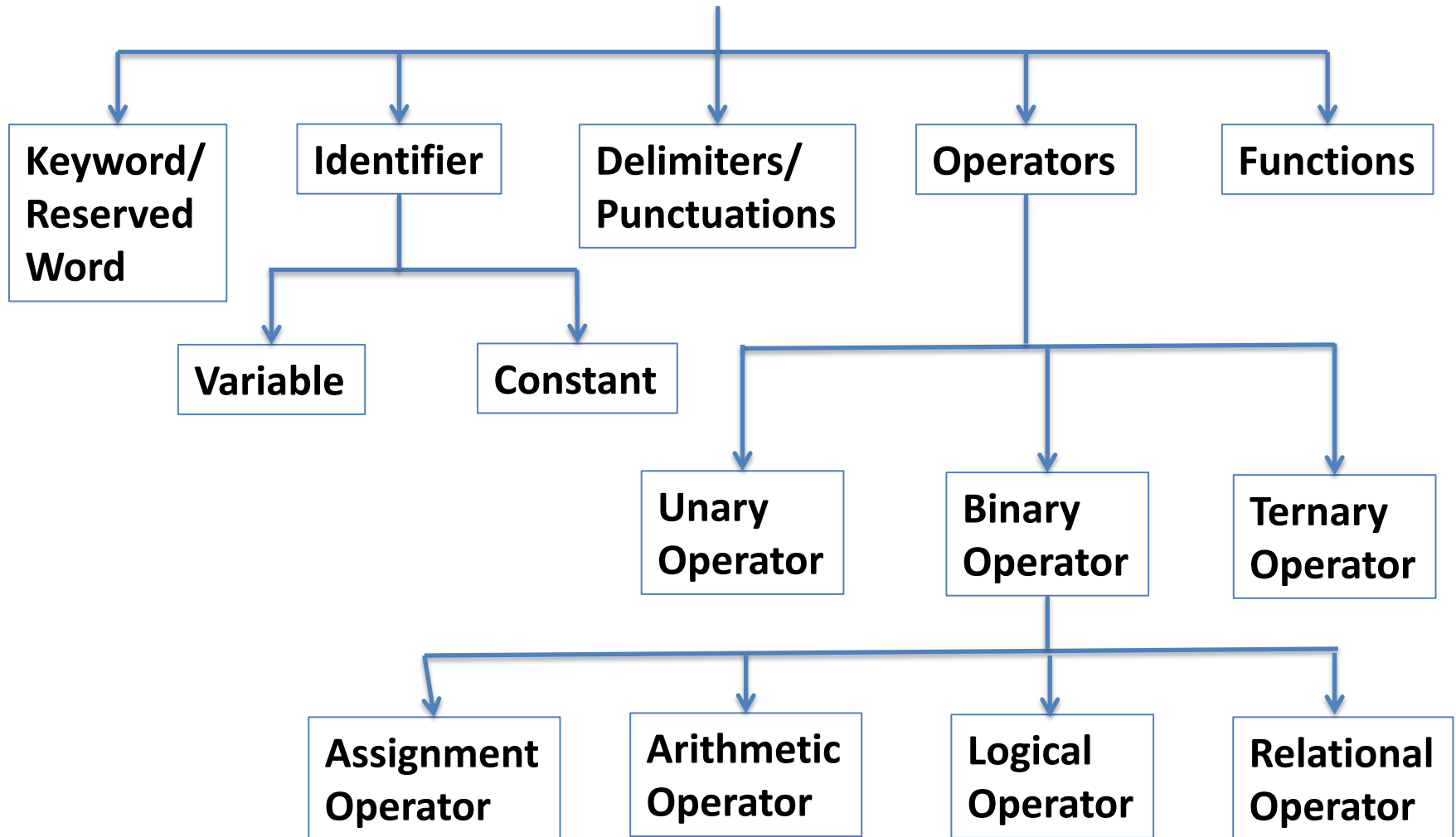
- ❖ There are some unprintable characters, which are followed by a backslash (\) and performs some specific tasks, are known as **escape sequence**.
- ❖ Some escape sequence with their details are given below:

Escape Sequence	Character	ASCII Code
\0	NULL Character	0
\a	Bell	7
\b	Backspace	8
\t	Horizontal Tab	9
\v	Vertical Tab	11
\n	Newline	10
\r	Carriage return	13
\"	Quotation Mark	34
\'	Apostrophe	39
\?	Question	63
\\	Backslash	92

~~Token~~

Classification of Tokens

Token



Keyword/Reserved Word

- ❖ **Keyword:** mandatory words to learn a language.
- ❖ **Reserved word:** words whose meaning or purpose is fixed and cannot be used for any other purposes.
- ❖ In any programming language, keywords and reserved words are the same set of words.
- ❖ Programming language is easy to learn because of few reserved/key words.

Some reserved/key words in C programming

include
for
case
long
void
const
return
Sizeof

define
do
break
char
auto
default
short
struct

main
while
continue
float
extern
else
signed
union

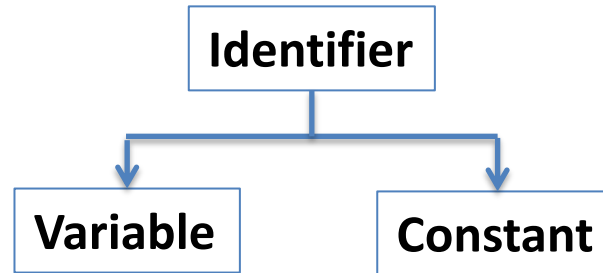
if
switch
int
double
static
goto
unsigned
typedef

Identifier

❖ **Identifier:** Identifier is a name given to a specific memory location.

Identifiers are of two types:

- ✓ Variable
- ✓ Constant



❖ **Variable:** when the content of an identified memory location is allowed to be changed, then the identifier is called variable. Variables are defined as follows:

```
int A, B;
```

```
char ch;
```

❖ **Constant:** Content of memory location cannot be changed. Constants are defined using macros, like

```
#define pi 3.14159
```

Identifier

Some Rules of defining Identifiers:

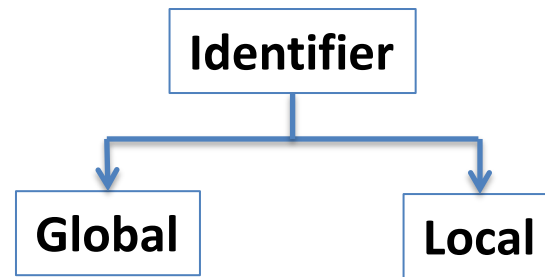
- Must be started with a letter or underscore (_).
- No space, delimiter or operator is allowed within an identifier.
- Should not be reserved/key word.

Different types of Identifiers:

- **char**
- Integer (**short**, **int**, **long**, **signed**, **unsigned**)
- Floating points (**float**, **double**)

Scope of Identifier

Classification of variable/Identifier based on scope:



- ❖ **Global Variable:** A variable declared before the `main()` function is called global variable. This variable can be accessed from every function of the program.
- ❖ **Local Variable:** A variable declared within any function can be accessed from that function only. This variable cannot be accessed from any other function. This type of variable is called local variable.

Scope of Identifier

```
#include <stdio.h>

void add(int x, int y);

int sum;

void main(){
    int x = 5, y = 7;

    add(x, y);
    printf("The sum is %d.", sum);
}

void add(int p, int q){
    sum = p + q;
}
```

Delimiters/Punctuations

❖ **Delimiter:** A delimiter is a sequence of one or more characters that specifies the boundary or acts as a separator.

Common delimiters used in C programming are-

- ✓ comma (,)
- ✓ Semicolon (;)
- ✓ Quotation (“ ”)
- ✓ Apostrophe (‘ ’)
- ✓ Parentheses (), Braces {} and Brackets []
- ✓ Forward slash (/), backslash (\)